

NeurIPS 2023 | SiT: Teaching Robots From the Middle of the Crowd

A dataset from Hanyang University captures pedestrian trajectories as a mobile robot navigates real crowds in downtown Seoul, with time-synchronized multi-modal sensors, semantic maps, and a unified perception-to-prediction benchmark.

For a robot to cross a busy hallway or a crosswalk safely, it has to treat the people around it as individual objects in 3D space and predict where each of them will move next. That sounds like a solved problem borrowed from self-driving cars, but the two situations are not the same. A car and the pedestrians it sees usually occupy separate lanes and sidewalks and rarely come into close contact. A delivery robot threading through a mall or waiting at a crossing is shoulder to shoulder with people the entire time.

The trouble is that the data we train these models on barely reflects the robot's actual situation. The classic pedestrian trajectory datasets — ETH, UCY, SDD — were filmed by fixed cameras mounted on rooftops or drones, so they never see a person react to a robot at all. Large driving datasets like nuScenes, Waymo Open, and Argoverse2 carry rich sensor data, but their pedestrians are typically off to the side on a separate path. Both settings miss the same thing: the close-range, space-sharing human-robot interaction (HRI) that a navigating robot lives with.

The answer this NeurIPS 2023 paper offers is the SiT dataset, short for Socially Interactive Pedestrian Trajectory. It is not another batch of overhead video. It was recorded by a mobile robot actually driving through indoor and outdoor crowds in downtown Seoul, and it ships synchronized LiDAR, camera, IMU, and RTK raw data, 2D and 3D annotations, multi-layered semantic maps, and a single benchmark spanning 3D detection, tracking, trajectory prediction, and end-to-end motion forecasting.

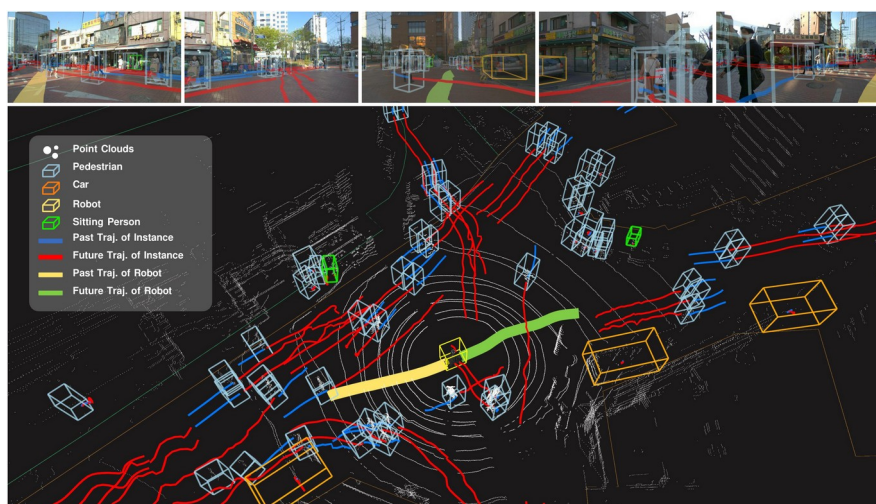


Figure 1. A visualization of one outdoor SiT scene (Cafe_Street_3). The top row is the 360-degree view stitched from five cameras. In the point cloud below, yellow is the robot's past trajectory and green its future

path, while blue and red trace each pedestrian's past and future motion; boxes are per-object 3D bounding boxes. The robot sits in the middle of the crowd, sharing the same space as the people around it.

Paper:

https://papers.nips.cc/paper_files/paper/2023/hash/4d6a000c216974f59e597bc878cd6325-Abstract-Datasets_and_Benchmarks.html

Code and development kit: <https://github.com/SPALaboratory/SiT-Dataset>

Why the existing datasets fall short

Work on human-robot interaction has long observed that a robot's motion changes how nearby people walk, and that the effect is strongest when the robot and the pedestrians share the same space at close range. To study and model that behavior, the training data has to come from a robot that is genuinely moving through a crowd — not from a camera bolted to a wall.

A few earlier robot-collected datasets came close, but each had a gap. STCrowd kept its sensors at a fixed position, so its scenes barely varied. JRDB was recorded in interactive scenarios, but it did not arrange its data as trajectories and did not fully time-synchronize its sensors, which limits sensor fusion. SiT is designed specifically to close those gaps.



Figure 3. Four representative SiT collection settings: an indoor hallway, a cafeteria street, an outdoor alley, and a crosswalk. In every shot the robot is surrounded by pedestrians, underscoring that the data was gathered inside real, dense foot traffic.

How the data was collected

The collection platform is a remotely operated Clearpath Husky unmanned ground vehicle. It carries two 16-channel Velodyne VLP-16 LiDARs, five Basler cameras that together cover a full 360 degrees, two IMUs, and RTK positioning. Fusing camera and LiDAR reliably depends on getting their timestamps to line up, so SiT triggers every sensor from a single centralized pulse-per-second (PPS) signal generator, keeping all the streams precisely aligned in time.

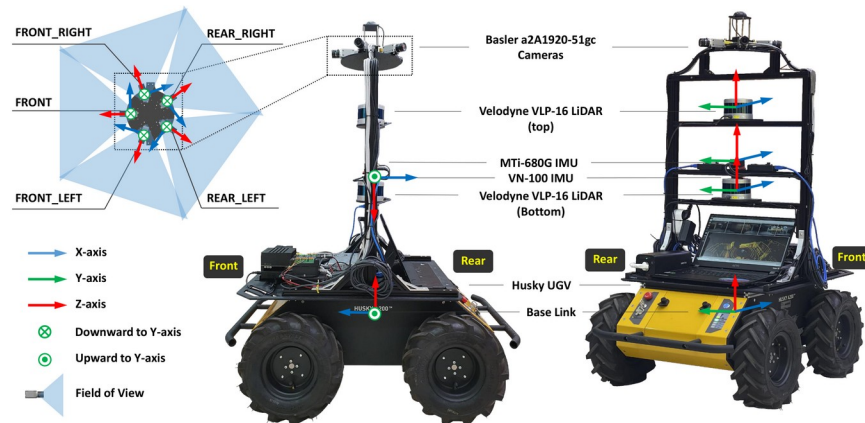


Figure 2. The sensor layout on the Husky UGV platform: five top-mounted Basler cameras give a 360-degree field of view (the inset at top left shows how they point), two Velodyne VLP-16 LiDARs sit at the top and bottom, and there are two IMUs (MTi-680G and VN-100) plus RTK. Colored arrows mark each sensor's coordinate axes.

The robot's own pose comes from RTK outdoors and from a LiDAR-inertial SLAM algorithm indoors. That step matters because the robot's ego-motion has to be compensated before pedestrian trajectories can be assembled correctly and the semantic maps can be aligned. Those maps use a 12-layer hierarchical structure, built from the point clouds and then segmented by hand. Expert annotators labeled 3D cuboids at 5 Hz, which were interpolated up to 10 Hz, and 2D boxes were derived from the 3D cuboids that share an object ID. Finally, for privacy reasons, every human face and vehicle license plate in the released data is automatically blurred.

The scale, and four benchmarks

In total SiT contains 60 scenes, which comes to roughly 60K images and 12K point cloud frames, carrying about 470K 2D annotations and 320K 3D annotations. Each clip is 20 seconds of sequential data sampled at 10 Hz, yielding 9 seconds of trajectory represented as pose vectors. The annotated classes go beyond pedestrians to include car, bus, truck, cyclist, and motorcyclist.

On top of the data, the paper defines four benchmark tasks: 3D pedestrian detection (scored with distance-based AP at 0.25/0.5/1.0/2.0 m thresholds), 3D tracking (standard metrics such as sAMOTA, AMOTA, and MOTA), trajectory prediction (2 seconds of past motion in, 7 seconds of future out, scored with ADE and FDE over the best of K candidates), and an end-to-end task that runs from raw sensors all the way to future boxes and trajectories.

The main finding: SiT captures the close interaction driving data misses

Does SiT actually capture the interaction it set out to? Plotting where pedestrians appear relative to the robot as a kernel density estimate (KDE) makes the answer easy to read.

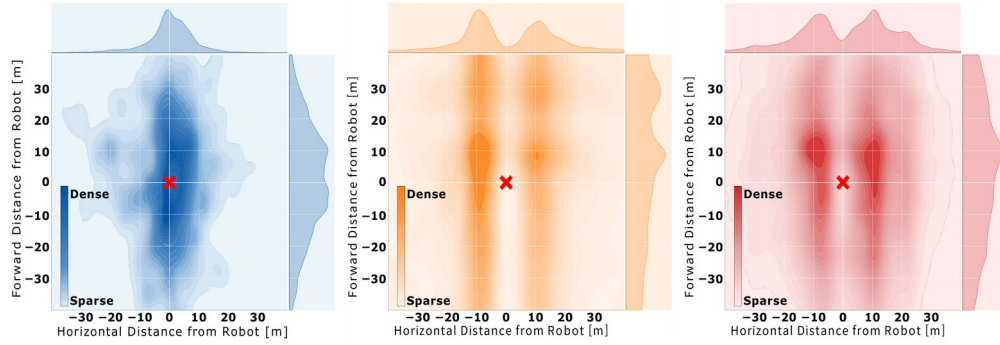


Figure 7. The spatial distribution of pedestrians relative to the ego body across three datasets (kernel density estimates; the red cross marks the robot or vehicle center). Left is SiT, middle Waymo Open, right nuScenes. In SiT the pedestrians cluster in close on all sides; in the two driving datasets they split off to the left and right, matching a scene where sidewalks and roadways are separate.

The quantitative version is just as clear. If you count the instances that satisfy both a space-sharing condition (robot and pedestrian in the same semantic space) and a proximity condition (within 2 meters of each other), SiT has far more than nuScenes, Waymo Open, or Argoverse2. In other words, driving data badly underrepresents exactly the close human-robot interaction a social navigation robot has to handle.

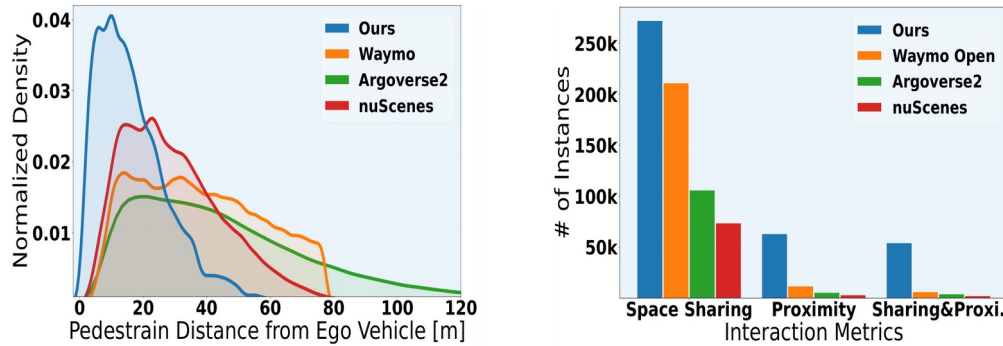


Figure 8. Comparison with autonomous-vehicle datasets. Left: the normalized density of pedestrian distance from the ego body — SiT (blue) is clearly higher at short range. Right: the instance counts satisfying the space-sharing, proximity, and combined conditions; on every one, SiT (blue) leads the other datasets by a wide margin.

The semantic map really does help prediction

SiT's semantic maps are more than a nice extra. Feeding a map to a trajectory-prediction model as scene context consistently improves accuracy. For Y-Net, adding the map lowers ADE20 from 0.836 to 0.675 and FDE20 from 1.878 to 1.547. For the strongest model, NSP-SFM, it brings ADE20 from 0.634 to 0.517 and FDE20 from 1.087 to 0.925. Both models benefit, which is direct evidence for the value of the scene context SiT provides.

Table 1. Effect of the semantic map on trajectory prediction (ADE20 / FDE20, lower is better; 2 s of input, 7 s predicted). Adding the map improves both models consistently, and NSP-SFM with the map is the best overall.

Model	No map ADE20 / FDE20	With map ADE20 / FDE20
Y-Net	0.836 / 1.878	0.675 / 1.547

NSP-SFM	0.634 / 1.087	0.517 / 0.925
---------	---------------	---------------

On the detection side the pattern is equally tidy: models that fuse camera and LiDAR beat single-modality ones. TransFusion with a voxel backbone reaches 0.531 mAP, ahead of CenterPoint at 0.518 and the camera-only FCOS3D at 0.244, and voxel backbones (SECOND) outperform pillar-based (PointPillar) ones. On tracking, the CenterPoint tracker leads with an sAMOTA of 0.6070 and an AMOTA of 0.2007.

Takeaway, and the honest boundary

The takeaway is straightforward. If you want robots that move safely and gracefully among people, you need training data that shows the crowd up close from the robot's own vantage point, with all the sensors aligned in time and the surrounding scene recorded as a map. SiT is the first dataset to bring those pieces together — collected by a robot inside real crowds, fully multi-sensor synchronized, paired with semantic maps, and backed by a benchmark that spans detection, tracking, prediction, and end-to-end forecasting.

It is worth being clear about the boundary. SiT was collected in indoor and outdoor settings in downtown Seoul, so its pedestrian behavior, street layouts, and cultural norms are regional; how well it transfers to other cities or crowds is an open question. That said, the dataset, the development kit, and the trained baselines are all publicly released, so the community can build on them — which is exactly what a benchmark dataset should make possible.